

Synthetic Conversation: How Computing Researchers Engage Multi-Perspective Dialogues to Brainstorm Societal Impacts

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Abstract

There have been increasing calls for computing researchers to consider the negative societal impacts of their work. However, anticipating these impacts remains challenging, as computing researchers often do not have access to diverse perspectives. Here, we explore how computing researchers brainstorm the negative societal impacts of computing innovations using a multi-agent dialog prototype called *WEAVER* as a probe that represents synthetic stakeholder conversations. Through think-aloud sessions and interviews with 12 participants, we evaluate how computing researchers perceive and engage with such a system, and whether they find it beneficial compared to using no such system or using ChatGPT. Our findings revealed that participants valued the conversations with the multi-agent system and considered societal impacts from new angles. Participants reported gaining insights and agency to reflect on issues rather than passively consuming pre-generated content. We discuss the findings, implications, and next steps for using a multi-agent system to brainstorm about the societal impacts of computing technologies.

CCS Concepts

• Human-centered computing → Interaction design.

Keywords

Multi-agent systems, Brainstorming, Human-AI Interaction, LLM

ACM Reference Format:

Rock Yuren Pang, Rohit Maheshwari, Julie Yu, and Katharina Reinecke. 2025. Synthetic Conversation: How Computing Researchers Engage Multi-Perspective Dialogues to Brainstorm Societal Impacts. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3719747>

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719747>

1 Introduction

In recent years, there have been growing calls for computing researchers to anticipate and consider the undesirable impacts of their work [20], e.g., in grant proposals [3] or paper submissions [14, 24, 33]. However, anticipating these impacts remains challenging. Although researchers are eager to consider undesirable impacts proactively, they often lack well-formulated processes and tools to do so effectively [11]. When doing so, even motivated practitioners can fail to envision downstream consequences for diverse stakeholders due to narrow perspectives [38, 43] and skewed demographic backgrounds [13, 32]. Considering and engaging with diverse and opposing viewpoints is crucial for overcoming the “failures of imagination” that often arise during the development of computing research systems [4].

Recent advancements in large language models (LLMs) offer a promising approach to address the challenge of limited perspectives in anticipating societal impacts. For instance, LLMs have demonstrated capabilities in enabling divergent thinking and ideation in contexts such as rapid idea generation [10], bursting online filtering bubbles [52], and facilitating group ideation [41]. A growing body of work in HCI has begun to prompt a single LLM to ideate potential harms in technologies, particularly in the fast-growing AI field [5, 34, 47]. On the other hand, studies have explored using multiple LLMs that interact with each other to generate more diverse and realistic simulations of human interactions in complex social settings [35] to anticipate future events [1, 16, 36, 50]. However, there is little understanding of how users perceive conversations among multiple synthetic agents and whether it may be engaging and useful for representing the *interplay* of diverse viewpoints when brainstorming about societal impacts.

This paper explores how users perceive interacting with a multi-agent dialog system, which consists of multiple LLMs that represent diverse viewpoints, in the context of anticipating the negative societal impacts of their projects. Our main research questions include: How do users perceive interacting with the multi-agent dialog system with synthetic conversation of diverse perspectives, when anticipating societal impacts? How do users value this interaction compared to other common approaches, e.g., brainstorming alone and using ChatGPT? We attempted to answer these questions using

a prototype called *WEAVER* as a probe. Insights into these questions may guide future system contribution to anticipate societal impacts.

We invited 12 computing researchers to explore the societal impact of their project. We captured the participants' experiences through think-aloud sessions and interviews, which allowed us to gain an in-depth understanding of their perceptions while engaging with LLM-simulated multi-perspective dialogues. Our findings show that participants value engaging in discussion with multiple stakeholders simulated by LLMs and that this approach broadened their perspectives and ideas around the societal impacts of their computing innovations compared to brainstorming without such a system or ChatGPT. Participants felt that synthetic conversation between multiple agents encouraged a deeper engagement with different viewpoints and potential negative societal impacts, rather than relying on a "laundry list" of high-level summaries that was provided by ChatGPT. They also reported that the interplay between the agents encouraged them to critically reflect on their research. We reflect on these encouraging findings and discuss future research needed to safely and ethically integrate multi-agent dialog systems into the research process.

2 Related Work

Anticipating Societal Impact. Anticipating impacts helps researchers foresee failures and potential risks of technologies that have not yet been observed or that may occur in new contexts. Prior work proposes to consider the impact throughout research development, such as the project ideation phase [9, 11], the grant proposal [3] and the paper submission [33]. To anticipate societal impacts, researchers can rely on checklists [33], impact assessment template [9], and seek expert opinions [40]. For example, the Delphi method iteratively gathers expert opinions until experts reach a consensus [30]. Furthermore, as a form of speculative design, design fiction creates a fictional future world to think through sociotechnical issues that have relevance and implications for the present [29]. Another key approach is Value Sensitive Design, which uses an iterative design process to account for human values in technology designs [15]. Our work was inspired by the rich norms and methods in the HCI community to design technology responsibly. Our work explores a novel interaction to anticipate consequences by debating with multiple synthetic agents.

Using LLMs to Brainstorm Societal Impact. Several works have proposed to incorporate LLMs to brainstorm negative societal impacts during the early stage of research thanks to their generative capabilities. For example, Farsight [47] is an *in situ* interactive envisioning tool that generates text descriptions about potential use cases, stakeholders, and harms in a node-link diagram. AHA! [5] generates descriptions of possible harms for different stakeholders in the form of an Ethical Matrix, which is then used to prompt an LLM to generate examples of harms. BLIP [34] leverages LLMs to generate summaries of undesirable consequences using relevant online articles. These tools show that LLMs can be a promising tool for exposing users to reasonable harms for awareness. However, the tools' evaluation also shows that an overabundance of harms may discourage users from considering them since users may be overwhelmed with the negative future of their project [5, 34, 47]. It is also prone to "alarm fatigue" where non-actionable alarms

lead users to feel overburdened with excessive problems [28]. On the other hand, automation might cause overreliance by the users, making them "sidestep the task of examining the impact of their technologies or could disincentivize them from learning to consider the consequences of their technologies [5]."

To support engagement with multiple perspectives, we draw parallels from the recent advance in the LLM multi-agent concept, where multiple LLMs collaborate or debate with one another to solve complex real-world tasks [18, 35]. Under this framework, LLMs assume distinct roles as agents with differing viewpoints or capabilities, simulating complex real-world discussions [35]. In the multi-agent team, each LLM agent has the potential to simulate personas [39, 42, 44]. This multi-agent LLM framework has since been applied in contexts like societal simulations [35], software engineering [49], decision-making [22], and content moderation [36]. Multi-agent frameworks have also been shown to improve performance in dialogue generation and question answering by fostering diverse interactions among agents [6]. However, prior work in multi-agent LLMs focuses on optimizing the technical performance of a given task, or the believability of such interaction. We focus on how computing researchers *perceive* and *engage in* synthetic conversations among multiple LLM agents, given that they are aware that the stakeholders are LLMs. We are interested in whether the conversations may improve user engagement and lead to useful insights when researchers anticipate societal impacts.

3 Methods

To answer our research questions, we conducted a user study using a probe called *WEAVER*. As we explore the design space and challenges using LLMs in brainstorming societal impacts, we chose to build an LLM-backed prototype, rather than a Wizard of Oz study to achieve more ecologically valid research [51] and to more accurately understand potential limitations and opportunities of this approach from the side of the users and the LLM output.

3.1 Apparatus

The prototype shows a discussion table (Figure 1), and has four main components: the project description input field (see supplemental materials), the chat table view (Figure 1a-b), the consequence summary column (Figure 1c), and the message view option (Figure 1d). After the project is defined, *WEAVER* automatically generates a group of stakeholders. Users have the option to edit or remove the generated stakeholders and add their own desired stakeholders. Subsequently, users can engage in conversations with this customized group of agents through the playground interface. In our interface, users can ask questions to engage in the conversations through the text input Figure 1a. Alternatively, they can refer to a list of conversation starters, which are questions adapted from the Tarot Cards of Tech [17]. The prototype automatically summarize the conversations as new consequences occur. We iteratively designed this prototype with three researchers prior to the user study. The backend, prompts, implementation details are summarized in the supplemental materials.

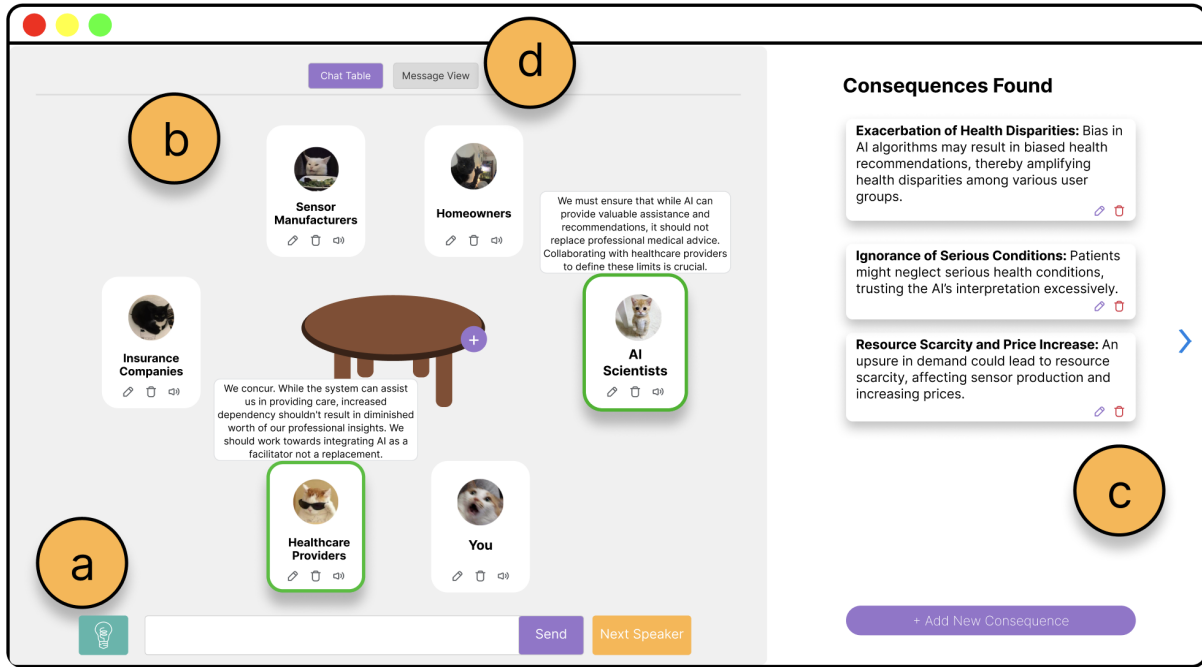


Figure 1: The prototype interface (WEAVER) includes the (1) playground for engaging in the synthetic conversation on the left; and (2) the consequence summary found based on the ongoing synthetic conversations on the right. Users can click the “Next Speaker” button or directly click on the stakeholder to continue the conversation.

3.2 Participants

All recruited participants ($N = 12$) had experience conducting research in computer science. They (four female and eight male, aged 23-29) are based in the US and are fluent in English. Our participant pool consisted of researchers from different areas of computer science research including ubiquitous computing, computer vision, natural language processing, video-language models, embodied AI, visualization, accessibility, social computing, and database management systems. All participants also interned or worked in industry relevant to their research fields. We met with three participants over Zoom and nine in person. Each participant anticipated the societal impact using their ongoing and unpublished work. To protect participants’ anonymity, we only refer to their general research directions and avoid discussing the project details and affiliations.

3.3 Procedure

Think-aloud sessions: First, participants were instructed to think of negative societal impacts without any assistance at first (BASELINE). During this condition, we provided a Google Doc template for them to note consequences and ways to prevent those consequences. To avoid confounds of typing speed and to lower the cognitive loads in the study, participants can choose to speak aloud with the authors and then summarize their thoughts after this condition. Second, participants used ChatGPT to further brainstorm consequences (CHATGPT). This helps us understand how the users’ experience might differ from simply prompting ChatGPT due to its prompt flexibility. To ensure the LLM quality consistency, we

provided access to the gpt-4o model in the ChatGPT web interface. Third, participants were asked to use our prototype to further anticipate impact (WEAVER). Before this condition, we gave a brief tutorial to demonstrate how to use it on an example of “Zoom video-conferencing.” This ensured participants had the chance to ask the researcher if anything was unclear. For each condition, participants were given 15 minutes to brainstorm and were instructed to think aloud. Participants were allowed to continue beyond the 15 minutes if they found the time insufficient. They can also stop in each condition if they feel they have run out of ideas or experience fatigue. To mitigate potential order effects, we counterbalance CHATGPT and WEAVER in our study to ensure our observation was not unduly affected by the knowledge or ideas in the previous conditions.

Exit Interview: The study ended with an interview where users provided qualitative feedback on interacting with multiple agents and potentially integrating the workflow into their research. We decided not to administer a post-study questionnaire to elicit subjective ratings on the system, based on early feedback in the pilot study to avoid response bias in favor of the prototype [8]. Instead, we encouraged comments and dialogues throughout the session.

3.4 Data Analysis

We recorded the participants’ screen and audio during the user study. We also noted when participants appeared to struggle and where interviewers intervened. We analyzed the data using a combination of affinity diagramming [2] and interpretation sessions [25]. Two authors independently reviewed each participant’s transcript and screen recording and distilled important insights from them.

All authors participated in reviewing and reorganizing the affinity diagrams in three interpretation sessions. We merged low-level emergent insights to synthesize observations across the interviews. We chose to use affinity diagramming as we aimed to discover opportunities and challenges for future use of systems like WEAVER or even simple ChatGPT, rather than building a comprehensive theory of current workflows. We also did not aim to compare them in a controlled lab experiment. As such, we followed a more practice-based approach, commonly used in HCI research [19, 21], to inform the design and application of new interaction methods for computing research when anticipating societal impact on their own.

4 Results

4.1 The presence of multiple stakeholders in the roundtable helps participants adopt diverse perspectives. Overall, participants indicated that the stakeholders effectively “*offered varied and distinct perspectives*” [P1]. P2 commented on the fact that “*WEAVER showed diverse stakeholders*”, which made it “*a lot easier [for me] to consider the different types of limitations or ethical and societal implications*.” All participants agreed that WEAVER presents stakeholders that they did not previously consider. Participants explained the reasons why the *presence* of diverse stakeholders is beneficial.

One reason is that **these stakeholders provide scaffolding to approach the difficult and abstract topic of negative societal impact**. The prototype helps users correlate the technology’s impact with people who might be using or involved with it in various capacities. For example, P4 stated that the presence of the diverse stakeholders made them consider questions such as “*what are the stakeholders? who are the stakeholders? are there more stakeholders?*” In fact, all participants checked the stakeholder definitions before the conversation started and made sure to revise the stakeholders to ensure relevance to their project. The presence of stakeholders inspired users to consider new tangent lines which already highlights new insights beyond their thoughts from **BASELINE**.

During a conversation, the presence of multiple stakeholders with titles and descriptions **established the credibilities of the societal impact discussed on the table**, as they transform abstract concepts into relatable perspectives. For example, P5 commented on the stakeholders that “*I see their role. They feel authentic, and they’re an expert*.” This gave the participant the impression that “*they are qualified to say this*.” However, many participants chose to agree or disagree with the stakeholder during the user study. P10 also noted that “*by using [WEAVER], it’s more like you know exactly what perspective the ideas are coming from*”. This is especially important because, in **CHATGPT**, participants said the insights from ChatGPT are too general and high-level in Section 4.2.

We also found that **the presence of stakeholders piqued their interest to add more stakeholders that were not initially present**. Of the 12 participants, 10 added new stakeholders after reviewing the initially generated stakeholders. For example, P4 works on a project that facilitates cross-domain communication between NLP and HCI. They mentioned that they have never considered the potential perspective from “*non-tech savvy individuals*”. Inspired by this idea, they continued to add another stakeholder — “*people who talk with a lot of jargon*” because they want to hear from the perspective of their potential users, who are different from

the stakeholders on the table. They stated that “*by adding a new stakeholder [to the] table, it gives me [information] specifically from the [stakeholder’s] perspective. That, I really like*.” After rounds of conversation, P8 noticed that although the stakeholders had different perspectives on many topics, they were similar on the topic of accessibility. Thus, P8 added a new stakeholder “*Someone who doesn’t care about accessibility*”. After asking the new stakeholder to speak, P8 agreed with what this new stakeholder had to say and was able to take away from the new, diverse perspective.

Another finding is that **the user interface, where the stakeholders are arranged around a central table, evoked a natural inclination for participants to engage in the conversation**. For instance, P6 noted that the arrangement of stakeholders around a table creates a “*group brainstorming vibe*,” which made them engaged in serious discussion. P6 further explained that this sense of discussion is important because it allows them to approach the issue without the pressure of right or wrong answers. Interestingly, we designed a message view (e.g., such as Messenger group chat) feature from an initial pilot feedback; however, all 12 participants chose the table view after they had initially explored the interface. The presence of multiple stakeholders in a roundtable might have encouraged a collaborative and reflective experience.

4.2 Participants gain actionable insights into their projects from the discussion and the summarization. Overall, participants believed that the conversations raised issues in the blind spots that could be relevant to their projects. For example, P9 worked on a project that tracks eye gaze for inside video meetings such as on Zoom. One synthetic stakeholder “*a speech-language pathologist*” stated that “*potentially those with strabismus or other eye alignment disorders could also be excluded, as the prototype might not correctly interpret or adjust their gaze*.” P9 commented that this is a completely “*new perspective that is worth looking into*.” P9 looked up this disease and told the authors that they would look more into this concern in their project. P12 stated that in **BASELINE**, he “*kind of [noted down] misinformation and data literacy*” as consequences but through WEAVER, they ended up adding “*creative exploration*,” which they believed really applied to their project. When considering the takeaways from the conversation, P12 also noted that some strategies have potential tradeoffs, and it’s important to determine whether “*the pros outweigh the cons*.” In another instance, P10 was building a project “*to provide a cost-efficient framework that will provide good [LLM] performance*.” When the “*open-source developers*” stated that users would lose trust if the framework was not well documented, P10 began to “*consider documentation as a strategy*,” and how “*the problem it is addressing is accessibility, or how people access the framework that [he is] building*.” When asked if they received any actionable insights from WEAVER, P10 decided to adopt this strategy for the project right away, as they commented that “*this is such a common practice that can be so easily overlooked*.”

4.3 Engaging with simulated conversations inspires further idea development amongst participants. Participants stated that engaging in conversation helps them “*break down the ideas*” [P4], “*lead to deeper thinking*” [P8], and “*think of problems with [their] project that [they] did not realize before*” [P3] during the conversation. During the study, we observed that participants spent 15.34 minutes ($SD = 4.20$) when interacting with **WEAVER**, which

is more than the time in the other two conditions (**BASELINE**: $M = 4.28$ minutes, $SD = 2.01$; **CHATGPT**: $M = 5.82$ minutes, $SD = 2.61$).

Participants spending less time when solely relying on prior knowledge and prompting ChatGPT made sense because they need to generate their own ideas, as opposed to select and evaluate ideas presented by the stakeholders [46]. Despite a long time with **WEAVER**, participants did not seem to “lose [their] attention” [P7]. Notably, we instructed participants that we were going to alert them after the 15-minute mark. Within this time, they should feel free to use the tool. Seven participants passed this point and continued to engage in the conversation after we alerted them.

Participants indicated that the **synthetic conversation inspires users to branch off from the conversation and develop new ideas**. In P12’s study, most stakeholders agreed with each other until the “policy makers” disagreed with the entire conversation. They pointed out that the opposing opinion was very “surprising to me.” They wished they could see the interaction between a “researcher themselves, [and] also the policy maker and discuss [their disagreement],” which helped him branch off the conversation to consider new viewpoints of what these stakeholders had to say. P10 was able to “think in different directions”. For instance, P10 realized that “small tech businesses” would prefer documentation, and added “documentations and code bases” to their summary list. P5 similarly noted that they had to “think about like regulation, think about manufacturing,” areas that they had overlooked in the past. The different topics in the conversation “[were] really helpful.” In other words, beyond the diverse stakeholders themselves, the back-and-forth conversations created more tangents to the ideation process.

The roundtable discussion made participants feel that they were immersed in the discussion. This happens even when participants disagree with the stakeholders. During the post-study interview, P4 mentioned that the conversation helped them think about their project “in a critical way, even though I do not agree with [the ideas].” P4 felt that the fact that agents on the roundtable may disagree with each other improved their “critical thinking.” Before prompting the stakeholders with the new question, P12 mentioned that “I actually have some ideas of what they might say.” In another example, P11, who works in developing vision-audio-language models, focused on two particular stakeholders in the conversation – language teachers and content creators. The dialogue between the two included disagreement on the topic of enhancing or detracting from personal relationships. Through this back-and-forth conversation, P11 began anticipating how the two stakeholders would respond to the situations. For instance, through some conversations, the participant was able to correctly identify multiple times that “the language teacher will disagree” or “the content creator will disagree.” They stated that “[ChatGPT] does not give [the ideas] in a way that I would like to discuss back and forth.”

We observed that **participants often agreed, questioned, or critiqued the points raised by the stakeholders on the table**. As P10 read aloud the stakeholder’s comments, they were confused and wanted to “expand the conversation a bit further” so that the stakeholders could “further explain what they mean,” indicating P10’s engagement and desire to question the stakeholder to elaborate on their response. For example, P8 worked on a project that uses drones to scan and image indoor features. One agent said “Scaling up the project tenfold could also accentuate the likelihood of

drones unintentionally capturing private conversation, further infringing upon the privacy rights of building occupants.” They disagreed by stating that this is “a very nice point, although the drone flying is very loud, and usually we don’t record any sound for that thing [drones].”

4.4 Participants preferred engaging in a discussion than passively consuming a “laundry list” of insights. Although a few participants had thought of the negative societal impact of their work prior to the study, many of the participants struggled with prompting ChatGPT. P2-10 provided a description of their project and a short query, such as “what is the social impact (or consequences)?” During **CHATGPT**, they spent most of their time reading the text from ChatGPT after prompting once or twice. P11 was the only participant to follow up on the ChatGPT’s response by incorporating the response in their new prompts. In particular, they were interested in the consequence of copyright issues in the art industry because they work in video-language model development. After using ChatGPT, P12 claimed that “this is just a list of stuff that is not presented in kind of a debate format.” P12 mentioned that in **WEAVER**, “not only [do] you have different perspective[s], but also their interactions with each other.” After using **WEAVER** and ChatGPT, the participant felt like “the other stakeholders who are kind of disagreeing or agreeing over these important issues and providing their own perspectives” is missing from the “laundry list of stuff” given by ChatGPT.

4.5 WEAVER should be used in the broader academic structure, potentially with other real-world stakeholders. A few participants indicated that they might not be in the best position to consider the consequences or implement some changes identified in the conversation. For example, P7 states how “[the impacts] were all good to have, but researchers might have their own priorities. So some are to be considered, but others are either too difficult or not focused on. And, in fact, I think some of those problems you can’t really solve very easily.” P10 further elaborates “this is something that I can do. But if there are some strategies that have other people [than themselves] involved, then it’s gonna be very hard to actually implement.” P11 details that, despite liking the takeaways, “I won’t be able to [change anything]. It’s something that has to be done by the higher-ups [in] leadership.” Participants often feel that they are not in the right position to implement the strategies that **WEAVER** generates. This indicates that adopting a system like **WEAVER** requires sociotechnical approaches such as more incentives to do so.

5 Discussion and Future Work

Our study shows that researchers viewed positively the synthetic conversation with multiple LLM agents when brainstorming societal impact about their own projects. Our study also showed that the success of brainstorming societal impact lies not only in the system output but also in the engagement and depth of reflection by users during the process. This observation aligns with prior work in HCI, which refers to engagement as not only the attitude and emotion to an interface but also an individual’s voluntary efforts to “understand and master challenging tasks” [12, 45]. The finding also extends the rich research that shows that collaboration and

discourse with humans improve understanding and learning of complex concepts [26, 31]. Our study revealed that the interplay of the synthetic conversation from stakeholders enabled participants to contextualize the societal impact in nuanced back-and-forth discussions. In contrast, while ChatGPT can generate comprehensive lists of societal impacts, most of which “*can be true*” [P10], participants were unlikely to engage deeply with such lists and internalize these insights. In P12’s words, ChatGPT only provides a “*laundry list of stuff, which lacks the context and clarity of an idea.*” They preferred WEAVER because “*not only [do] you have different perspective[s], but also their interactions with each other.*” Engaging researchers in anticipating societal impacts is crucial because the nature of the task requires more than usual creative processes. The exposure to new perspectives can spark ideas [37, 52], but they alone may not uncover the deeper, often uncomfortable, negative consequences of their projects. This finding implies that future work designing tools to anticipate societal impact needs to focus on engaging researchers in this difficult task, rather than simply presenting this information.

One limitation of this work is the relatively limited sample size and diversity of participants in this study, due to the preliminary nature of this work. Our result may also be affected by response bias, a common issue in studies like ours. Given the limited time in the user study, participants might have developed more ideas about societal impact on their own or with ChatGPT. Due to novelty bias [27], participants might have favored the new interface more than others. However, we made it explicit at the beginning of our user study that our goal was to explore how synthetic conversations might complement or expand upon individual reflection or the use of existing tools like ChatGPT. Moreover, the time constraints in the study may have limited the depth of their engagement with the tool. Future work could allow for incorporating follow-up sessions after several projects to observe how perceptions of the synthetic conversations may evolve in practice. This could also explore translational resources [7] to put a tool like WEAVER into practice (section 4.5). Future work could also compare synthetic conversations to real-world conversations in anticipating social impact, though real-world brainstorming conversations have shown to have issues such as peer judgment, free riding, and production blocking [23].

To anticipate societal impact of our own project, we must acknowledge that our results and prototype design may incur misrepresentation and misuse. Although WEAVER may provide diverse and opposing perspectives, it is not portrayed to replace consulting with real stakeholders to anticipate the societal impact of researchers’ projects. LLMs can introduce serious issues such as model hallucination [48], leading to inaccurate or misleading information. To address this, we cautioned the participants to be aware they are conversing with different personas represented by gpt-4o. We asked participants to treat their experience in the user study as a starting point to continue reflecting on the societal impacts. Interestingly, while all participants were fully aware that the stakeholders and conversations were synthetic, many engaged in discussions beyond the allotted 15 minutes. Given that all participants had a high education level (i.e., all have or are pursuing a PhD degree), they could suspend disbelief to some extent and engage in the conversations. To avoid misuse, we implemented safeguards, e.g., disclaimers, and maintained a list of blacklisted stakeholders (e.g., misogynists) to prevent the system from generating inappropriate personas.

Acknowledgments

We thank our participants and the anonymous reviewers for their valuable feedback. We also thank Kevin Feng, Quan Ze Chen, Katelyn Mei for their helpful suggestions. This work was supported by the National Science Foundation under Grant No. 2315937. Any opinions, findings, and conclusions or recommendations expressed in this paper are those of the authors and do not necessarily reflect the views of the National Science Foundation.

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